Name:____________

Double Award Science – Applied Unit 3 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /75	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2018	15	17	21	25	30	35	40
2019	8	11	16	21	27	33	39
2022	9	12	17	22	27	31	35
2023	10	13	18	23	28	33	39

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3445UC0-1

S18-3445UC0-R1

APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes

HIGHER TIER

TUESDAY, 15 MAY 2018 – AFTERNOON

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	14	
2.	5	
3.	14	
4.	14	
5.	8	
6.	12	
7.	8	
Total	75	

3445UC01
01

ADDITIONAL MATERIALS

A calculator.

INSTRUCTIONS TO CANDIDATES

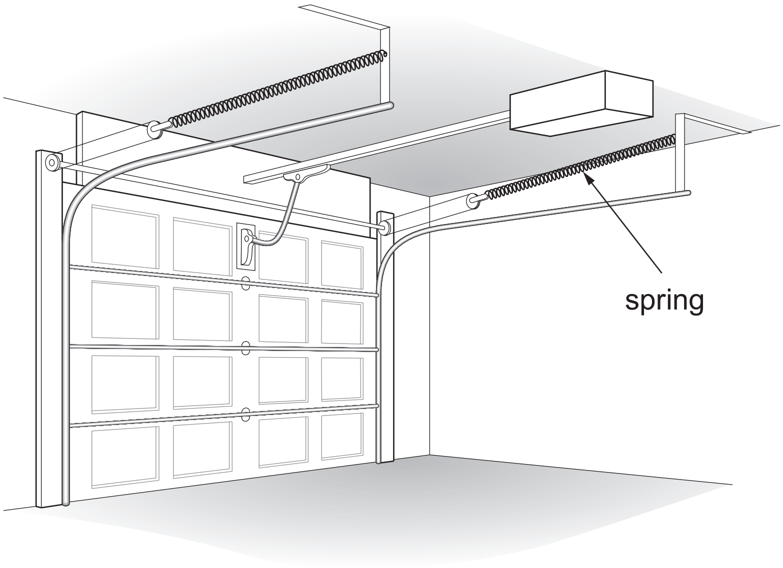
Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6**(*a*) is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of the examination page.

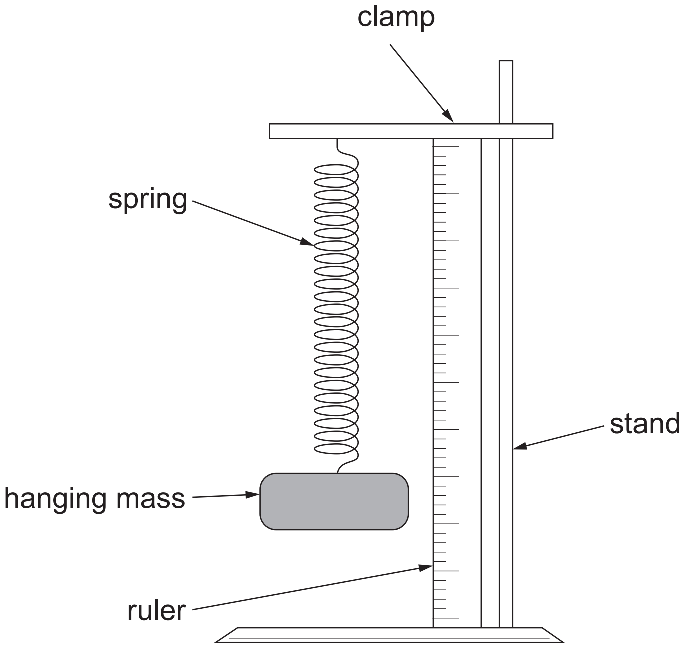
*Answer **all** the questions in the spaces provided.*

1. Springs can be used in garage doors. When the door shuts the springs are stretched and put under tension. When the door opens the spring returns to its original length making it easier to raise.

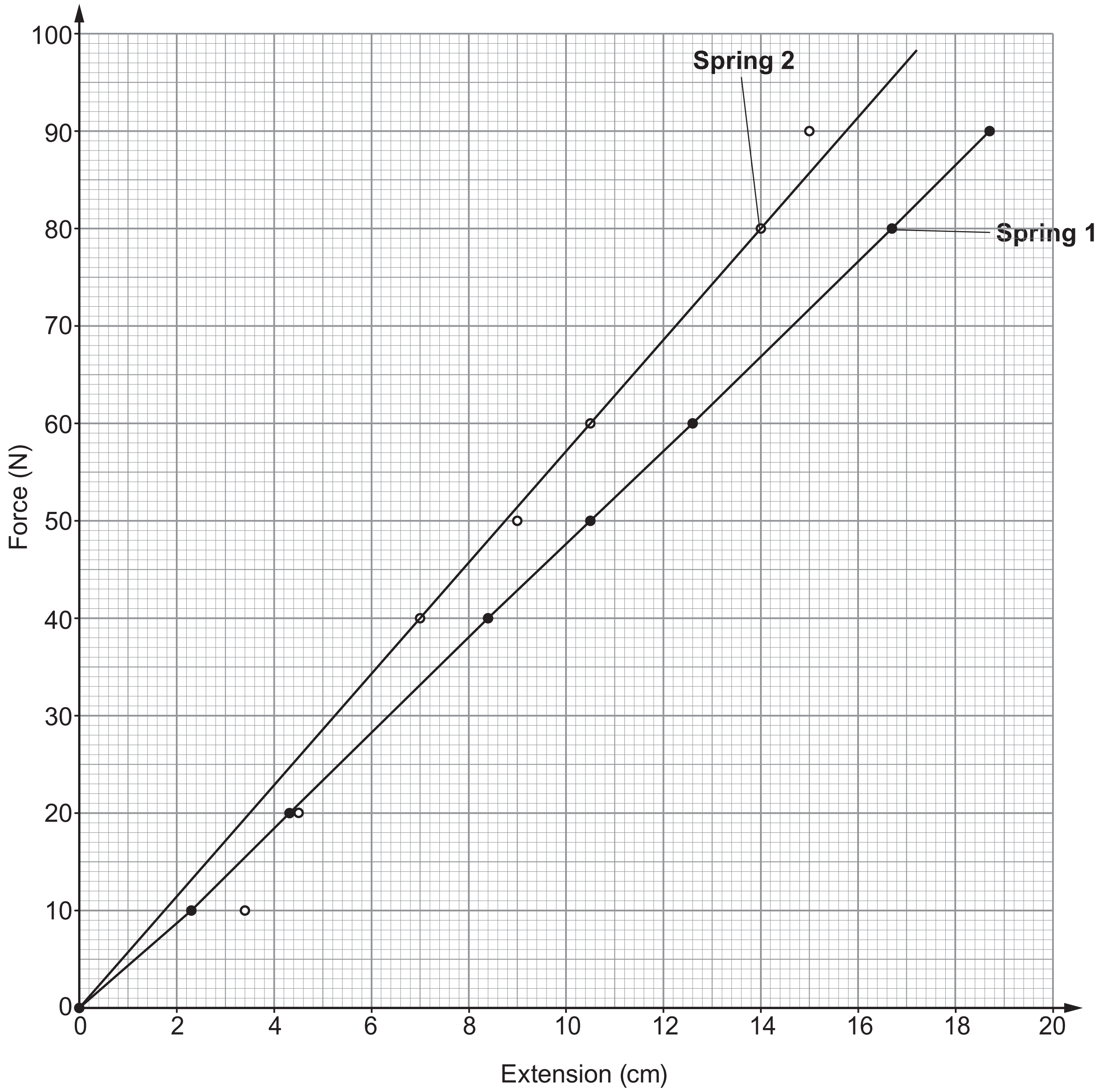


It is important that the spring is designed so that it does not become permanently stretched during the process. Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

Students test three springs using the apparatus shown below.



(a) Their results for two springs used in the doors are shown in the graph below.



Examiner only

(i) Compare the behaviour of the springs as the force increases from 10 N to 40 N. [2]

....

....

....

(ii) Use the graph and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the value of the spring constant for **Spring 1**. [3]

spring constant = ... N/cm

(iii) Use the graph to state **one** reason why the students can have more confidence in the results for **Spring 1** than for **Spring 2**. [1]

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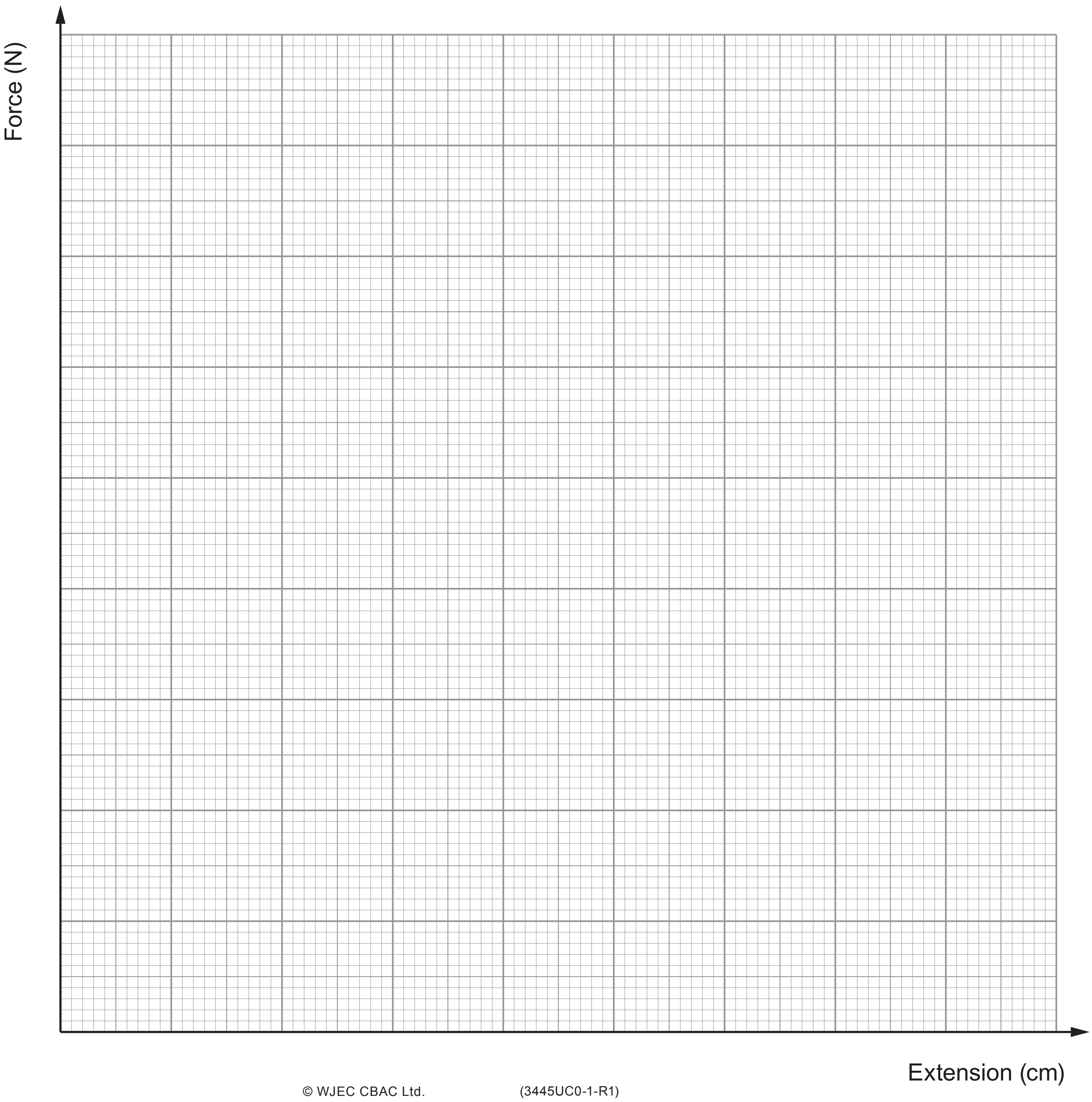
Examiner only

(b) The results for **Spring 3** are given in the table below.

Force (N)	Extension (cm)
0	0.0
10	2.4
20	4.8
40	9.6
50	12.0
60	16.0
80	28.0
90	36.0

(i) Plot the data on the grid below and draw a suitable line. [4]



Examiner only

(ii) Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

I. **Label** the elastic limit on your graph with a letter **E**. [1]

II. Use your graph to find the force required to reach the elastic limit. [1]

Force = ... N

(iii) During the opening and closing of the garage doors the spring will extend by 30 cm. Explain whether or not **Spring 3** is suitable for this purpose. [2]

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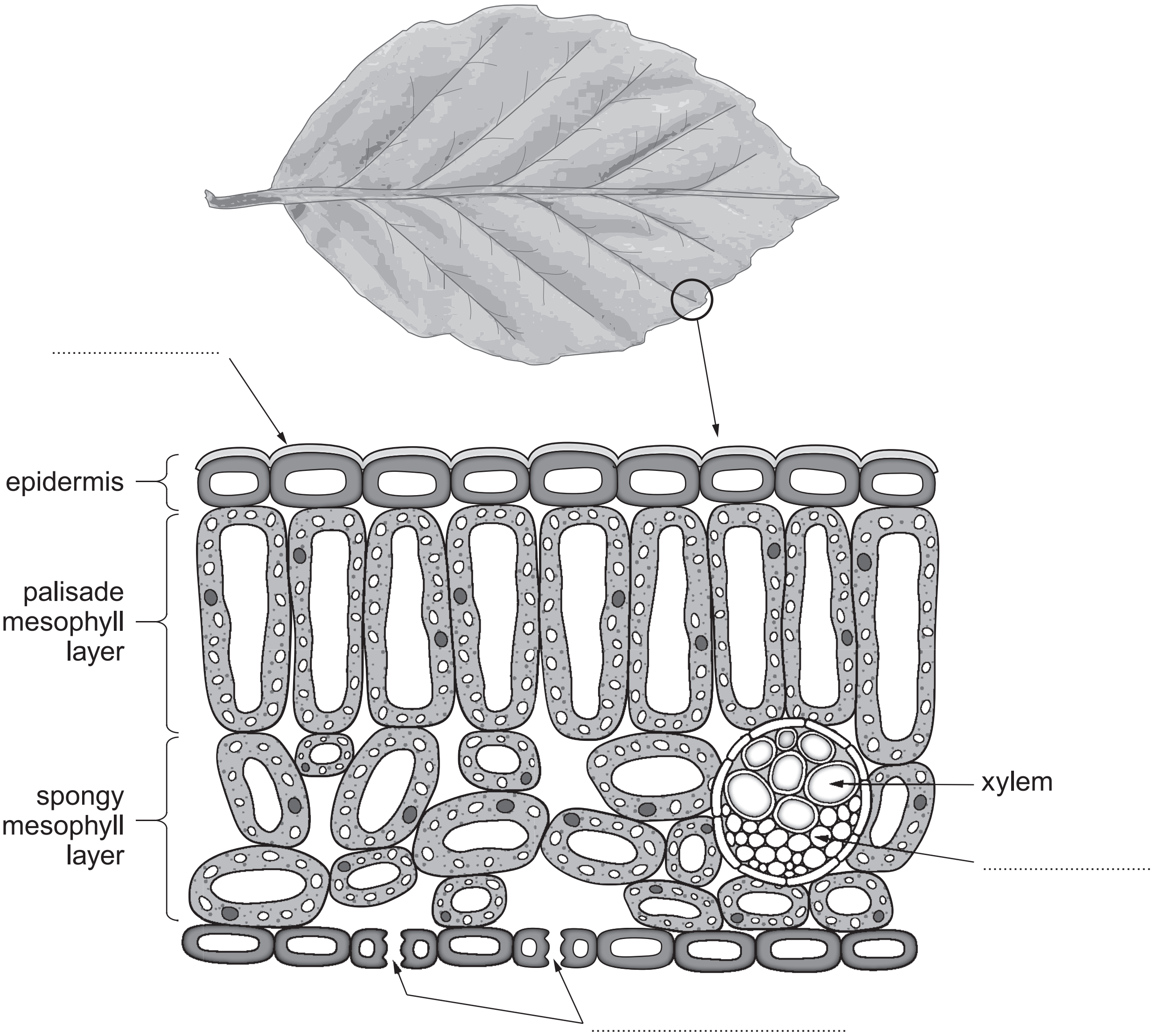
14

3445UC01 07

Examiner only

2. Students are learning about plants. They are interested in the structure and function of leaves.

(a) **Complete** the labelling of the parts of the leaf in the diagram below. [3]



(b) State the advantage of a transparent epidermis. [1]

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(c) State the purpose of the spongy mesophyll layer containing air spaces. [1]

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5

Examiner only

3. Metals have a variety of applications in industry, including making the wings and wiring for aeroplanes.

(a) (i) Describe metallic bonding. [2]

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(ii) Explain the property of metals which allows them to be used to make sheets for aeroplane wings. [2]

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(iii) Explain why the metals used in wires are good conductors of electricity. [2]

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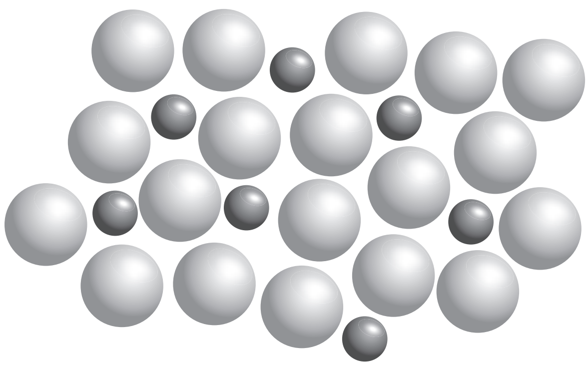
(b) Wings of modern jet fighter aircraft are made from alloy X which is composed of titanium, aluminium and vanadium. Wings on older aircraft were made from steel.

(i) Alloys consist of more than one type of atom, as shown in the diagram below.



Explain why this increases the strength. [2]

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Examiner only

(ii) Some properties of these metals and alloys are given in the table below.

Metal	Density (kg/m^3)	Stiffness (GPa)	Tensile strength ($\times 10^7$ Pa)
titanium	4500	110	55
aluminium	2700	69	10
vanadium	5700	138	63
alloy X	4400	110	100
steel	7800	210	40

I Compare the use of alloy X with steel for making aeroplane wings. [2]

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II The mass of a wing made from alloy X is 2500 kg. The alloy is 90% titanium and 6% aluminium.
Use the equation:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the volume of vanadium needed for each wing. [4]

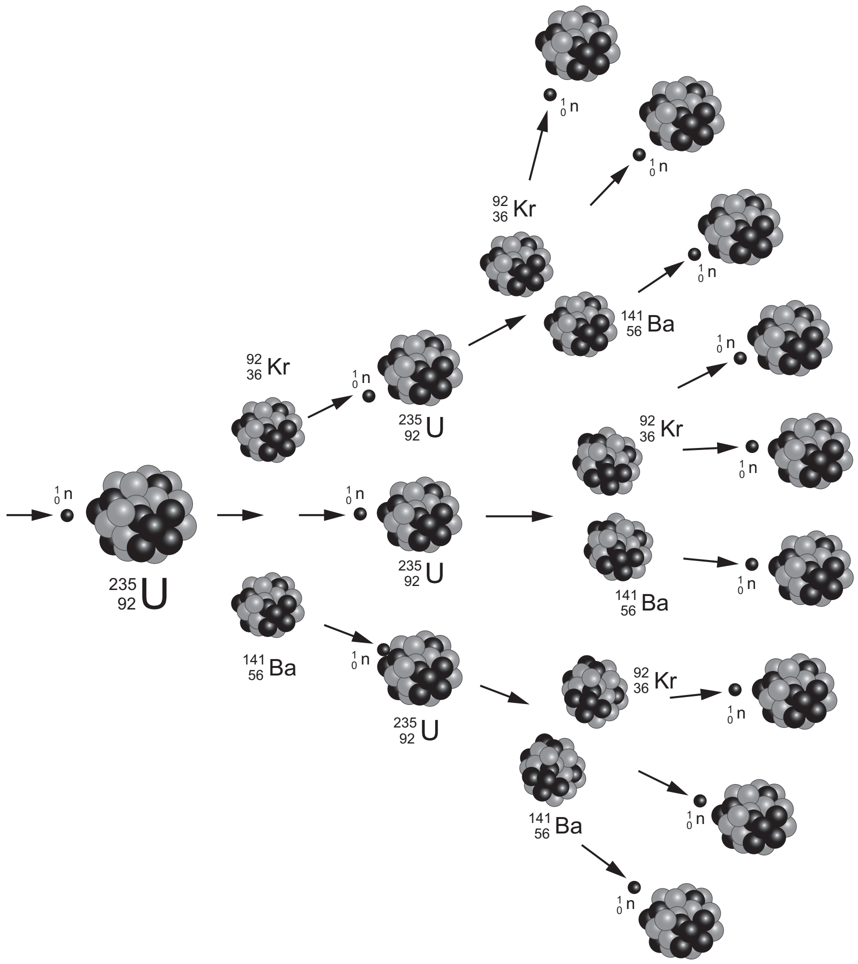
Volume = ... m^3

14

4. Information about two natural isotopes of uranium is given in the table.

Isotope	Symbol	Type of decay	Half life (years)	Product
uranium-235	$^{235}_{92}U$	alpha	7×10^8	thorium-231
uranium-238	$^{238}_{92}U$	alpha	4.5×10^9	thorium-234

One type of fission reaction involving uranium-235 is shown in the diagram.



Another type of reaction occurs in a fast breeder fission nuclear reactor. Uranium-238 (U-238) can be used to generate plutonium-239 (Pu-239), which also acts as a reactor fuel supply. An intermediate stage involves the production of neptunium-239 (Np-239).

The reaction follows the three stages below:

$$\text{U-238} + n \rightarrow \text{U-239}$$
$$\text{U-239} \rightarrow \text{Np-239} + \beta$$
$$\text{Np-239} \rightarrow \text{Pu-239} + \beta$$

Examiner only

(a) (i) State what is meant by the term half-life. [2]

....

....

(ii) The isotopes of uranium in the table have been in existence since the Earth was formed. One estimate of the Earth's age is 4.2 billion years. Calculate the fraction of uranium-235 that remains on Earth today. [3]

Fraction = ...

(b) (i) Explain how the uncontrolled chain reaction shown in the diagram is prevented in a nuclear reactor. [3]

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(ii) Describe the structure of a nucleus of ${}^{141}_{56}\mathrm{Ba}$ which is one of the fission products. [2]

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(c) Complete the following nuclear equations showing how uranium-238 generates plutonium. [4]

$${}^{238}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{239}_{\dots\dots}\mathrm{U}$$

$${}^{239}_{\dots\dots}\mathrm{U} \longrightarrow {}^{239}_{\dots\dots}\mathrm{Np} + {}^{\dots\dots}_{\dots\dots}\beta$$

$${}^{239}_{\dots\dots}\mathrm{Np} \longrightarrow {}^{\dots\dots}_{\dots\dots}\mathrm{Pu} + {}^{\dots\dots}_{\dots\dots}\beta$$

14

5. Forensic scientists analyse substances found at crime scenes. They use analytical techniques to arrive at qualitative conclusions. These techniques are being used by students to identify samples of an unknown powder.
The table below shows the effect of adding sodium hydroxide solution to selected cations.

Cation	Effect of adding sodium hydroxide solution
aluminium (Al^{3+})	white precipitate soluble in excess sodium hydroxide to give a colourless solution
ammonium (NH_4^+)	no precipitate when heated with sodium hydroxide, ammonia gas is given off
calcium (Ca^{2+})	white precipitate insoluble in excess sodium hydroxide
copper(II) (Cu^{2+})	blue precipitate insoluble in excess sodium hydroxide
iron(II) (Fe^{2+})	green precipitate insoluble in excess sodium hydroxide
iron(III) (Fe^{3+})	reddish-brown precipitate insoluble in excess sodium hydroxide
lead(II) (Pb^{2+})	white precipitate soluble in excess sodium hydroxide to give a colourless solution
zinc (Zn^{2+})	white precipitate soluble in excess sodium hydroxide to give a colourless solution
magnesium (Mg^{2+})	white precipitate insoluble in excess sodium hydroxide

Tests for some anions are shown in the table below.

Anion	Test	Result
carbonate (CO_3^{2-})	add hydrochloric acid	carbon dioxide gas is produced
chloride (Cl^-)	acidify with dilute nitric acid, then add silver nitrate solution	white precipitate formed
iodide (I^-)	acidify with dilute nitric acid, then add silver nitrate solution	yellow precipitate formed
nitrate (NO_3^-)	add sodium hydroxide solution, then aluminium and warm carefully	ammonia gas produced
sulfate (SO_4^{2-})	acidify with dilute hydrochloric acid and then add barium chloride solution	white precipitate formed
bromide (Br^-)	acidify with dilute nitric acid then add silver nitrate solution	cream precipitate formed

Examiner only

(a) Describe the difference between qualitative and quantitative analysis. [2]

....

....

(b) The students dissolve samples of the unknown powder in water and test for anions and cations.
The observations from the students' tests are shown below.

Test	Observations
sodium hydroxide solution added	white precipitate which did not dissolve as more sodium hydroxide was added
acidify then add barium chloride solution	white precipitate
hydrochloric acid added	bubbles of gas
dilute nitric acid then silver nitrate solution added	white precipitate

Use the information in the tables above and opposite to determine:

(i) all the possible cations that may be present. [1]

....

(ii) all the anions that may be present. [1]

....

(c) Describe another test that could be used to confirm the cation(s) present. [2]

....

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....

(d) Name **one** compound that could be present in the solution and use the symbols given in the tables to write down its chemical formula. [2]

Name ....

Formula ....

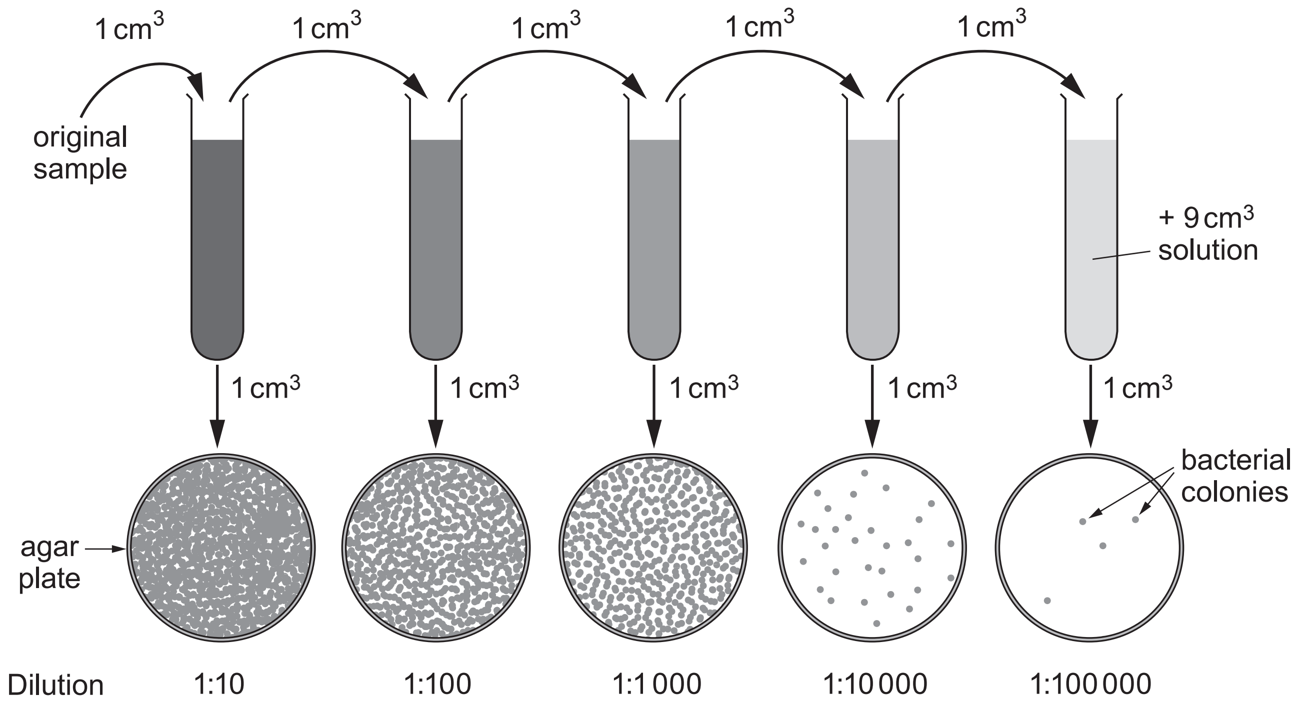
8

Examiner only

6. *(a)* Explain the methods used by the food industry to preserve food. [6 QER]

Examiner only

(b) Whenever an outbreak of food poisoning occurs, environmental health officers will carry out an investigation including testing samples of food for the presence of bacteria.
To ensure a countable agar plate a series of dilutions is carried out as shown in the diagram below. The number of bacterial colonies on each plate is counted.



(i) Suggest which dilution in the diagram above will provide the best set of results **and** give a reason for your answer. [2]

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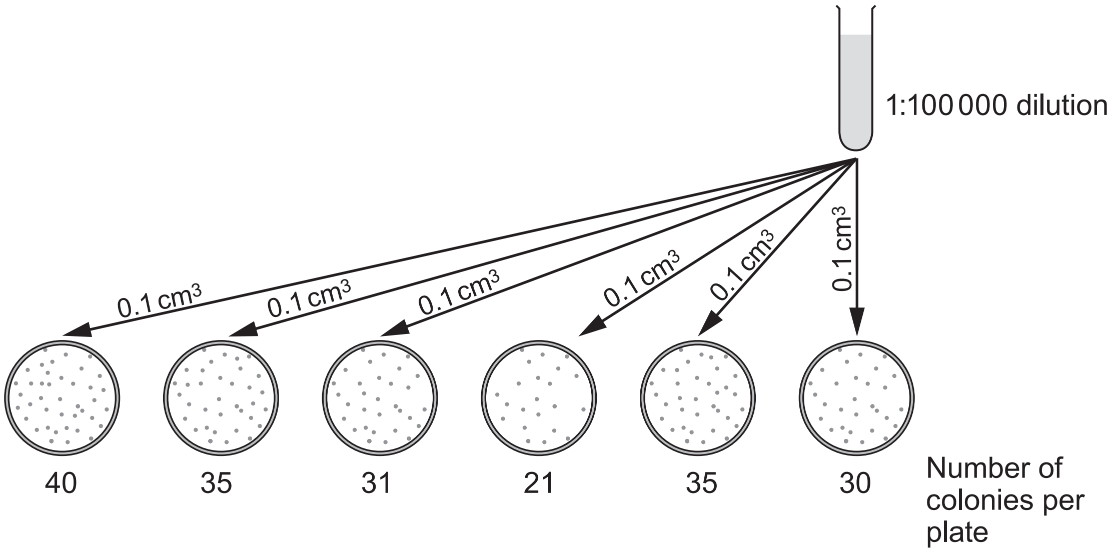
Examiner only

(ii) In another investigation, six agar plates were prepared by adding 0.1 cm^3 of solution from a 1:100 000 dilution. The agar plates were left for 72 hours. The results are shown below.



Use the information above to calculate the number of bacteria/cm^3 in the original **undiluted** sample. [4]

Number of bacteria = ... / cm^3

12

Examiner only

7. The chemical industry provides many of the chemicals that people need for modern life. The chemical industry today is developing new processes to manufacture these chemicals more efficiently and with less impact on the environment.

(a) Explain why catalysts allow chemical reactions to be carried out at lower temperatures than would otherwise be needed. [2]

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(b) Describe the economic and environmental benefits of developing new and better catalysts. [3]

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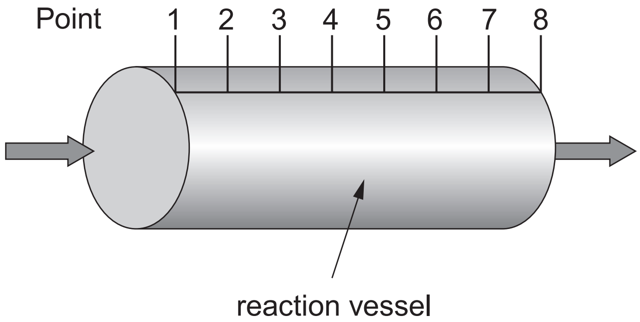
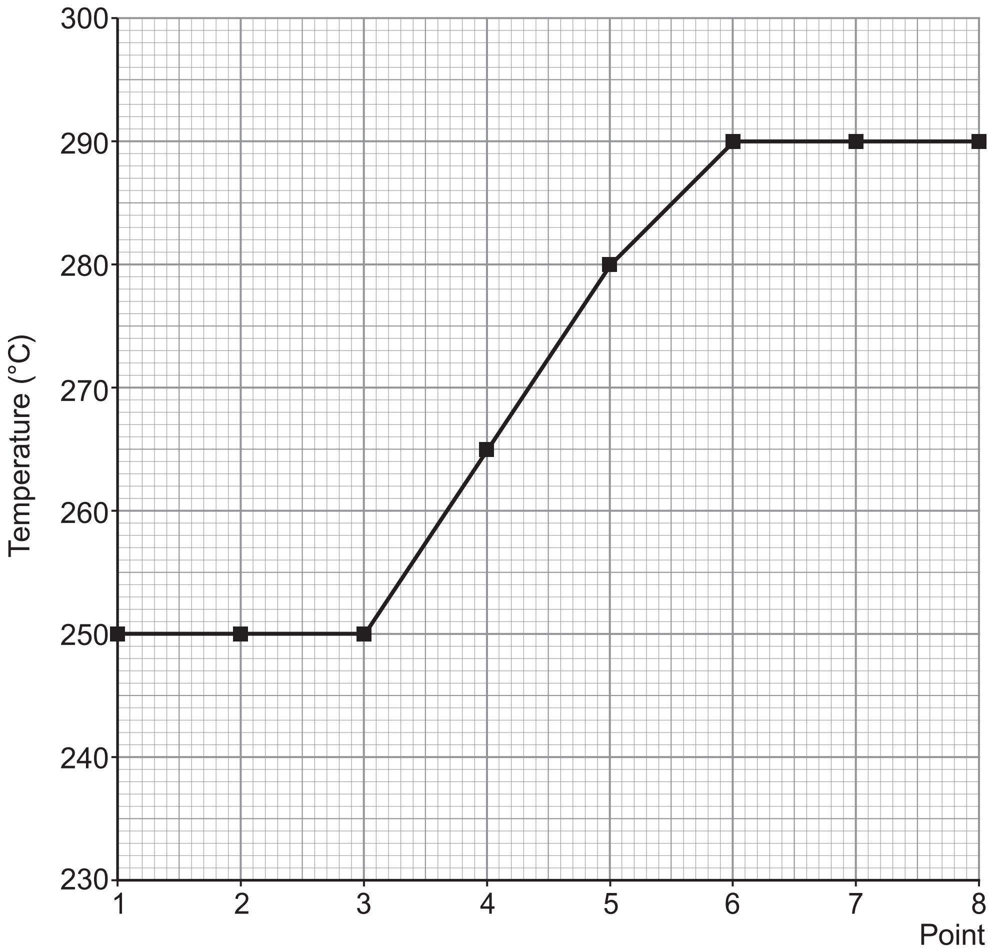
(c) Analysis of a reaction gives a good measurement of catalyst performance.



In an industrial process, waste gases, which include hydrogen sulfide, are pumped through a reaction vessel containing a catalyst. The hydrogen sulfide is removed in an **exothermic** reaction which causes an increase in gas temperature. The catalyst is performing badly if the hydrogen sulfide is not removed. Thermocouples monitor the temperature across a reaction vessel at eight different points as shown above.

Examiner only

The temperature at each of the points 1 to 8 in the reaction vessel are plotted on the graph below.



Evaluate the performance of the catalyst across the reaction vessel. [3]

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8

END OF PAPER

THE PERIODIC TABLE

1	2	Group										3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3445UC0-1

S19-3445UC0-1

TUESDAY, 14 MAY 2019 – AFTERNOON

APPLIED SCIENCE (Double Award)

UNIT 3: Food, Materials and Processes

HIGHER TIER

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**12**	
3.	**6**	
4.	**13**	
5.	**18**	
6.	**6**	
7.	**13**	
Total	**75**	

3445UC01 01

ADDITIONAL MATERIALS

A calculator.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **3** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of the examination paper.

Examiner only

Answer **all** *the questions in the spaces provided.*

1. Photosynthesis is the process by which green plants produce their own glucose.

(a) Complete the word equation for photosynthesis below. [1]

.... + ... $\longrightarrow$ glucose + oxygen

(b) State **three** different ways green plants use the glucose they produce. [3]

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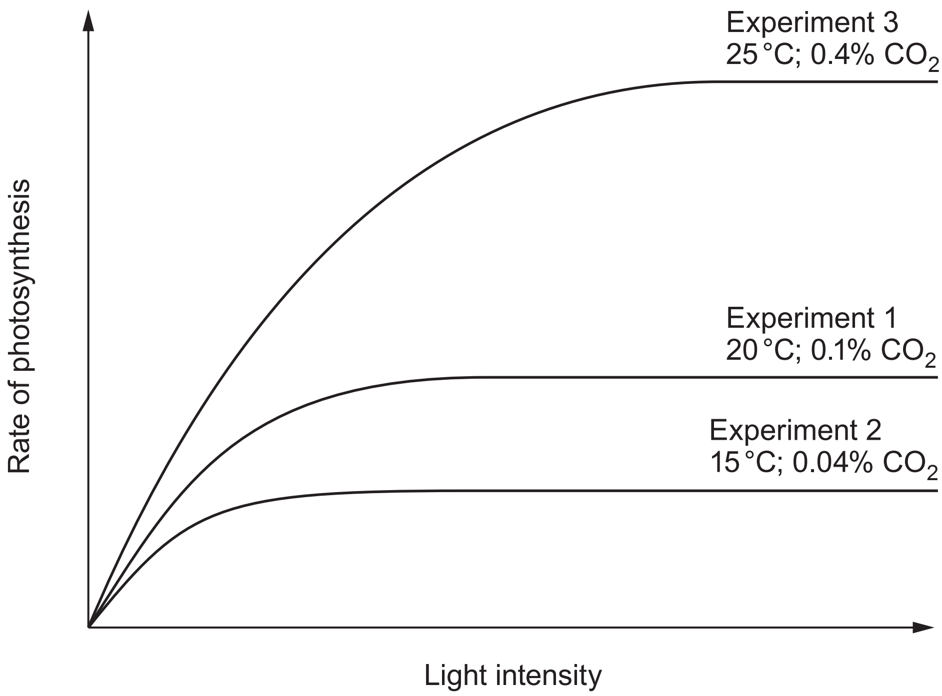
(c) The rate of photosynthesis is affected by light intensity, temperature and carbon dioxide concentration as shown in the graph below.



Use the information in the graph to describe how differences in light intensity, temperature and carbon dioxide concentration affect the rate of photosynthesis. [3]

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Examiner only

2. *(a)* Food poisoning is a notifiable disease in Wales. Doctors are required to report every case of food poisoning.

(i) Explain how some bacteria cause food poisoning. [2]

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(ii) State **two** symptoms of food poisoning. [1]

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(iii) State **two** different types of precaution that can be taken when **preparing** food to reduce the risk of food poisoning. [2]

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(b) **Figure 1** shows the number and rate of reported cases of food poisoning in Wales between 1992 and 2016.

Figure 1

Year	Total number of reported cases	Rate (number per 100 000 population)
1992	3 590	124.5
1998	5 946	205.0
2000	4 716	162.2
2006	4 301	145.4
2010	4 980	165.6
2014	4 516	146.1
2016	4 504	145.7

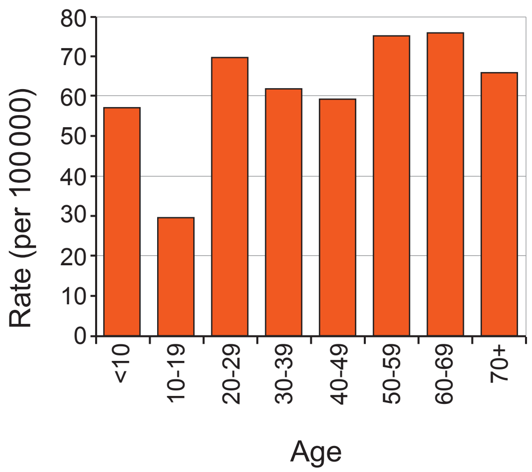
The percentage of reported cases of food poisoning during each quarter of a year is shown in **Figure 2**.

Figure 2

Quarter	% of total for year		
	2012	2014	2016
Jan-Mar	18	17	19
Apr-Jun	26	26	25
Jul-Sep	32	32	33
Oct-Dec	24	25	23

Figure 3 shows the rate of food poisoning amongst females by age group in 2016.

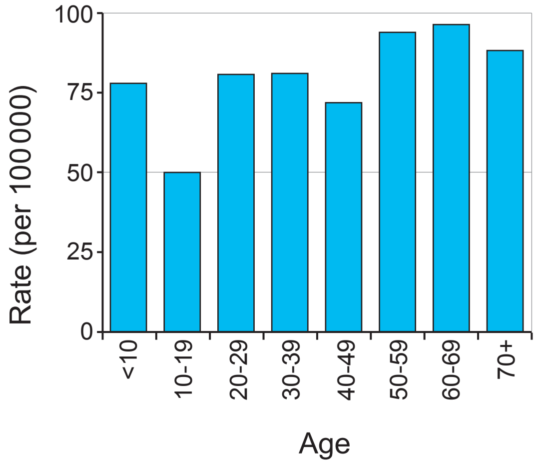
Figure 3



Examiner only

Figure 4 shows the rate of food poisoning amongst males by age group in 2016.

Figure 4



Use the data in **Figures 1 to 4** to answer the following questions.

(i) Since 1998, egg-laying hens have been vaccinated against a species of *Salmonella*. Explain how this has affected the number of reported cases of food poisoning. [2]

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(ii) Explain the variation in the percentage of reported cases of food poisoning in each of the quarters in a year. [3]

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(iii) It is suggested that the rate of food poisoning does not depend on gender or age. Explain whether this suggestion is confirmed by the data. [2]

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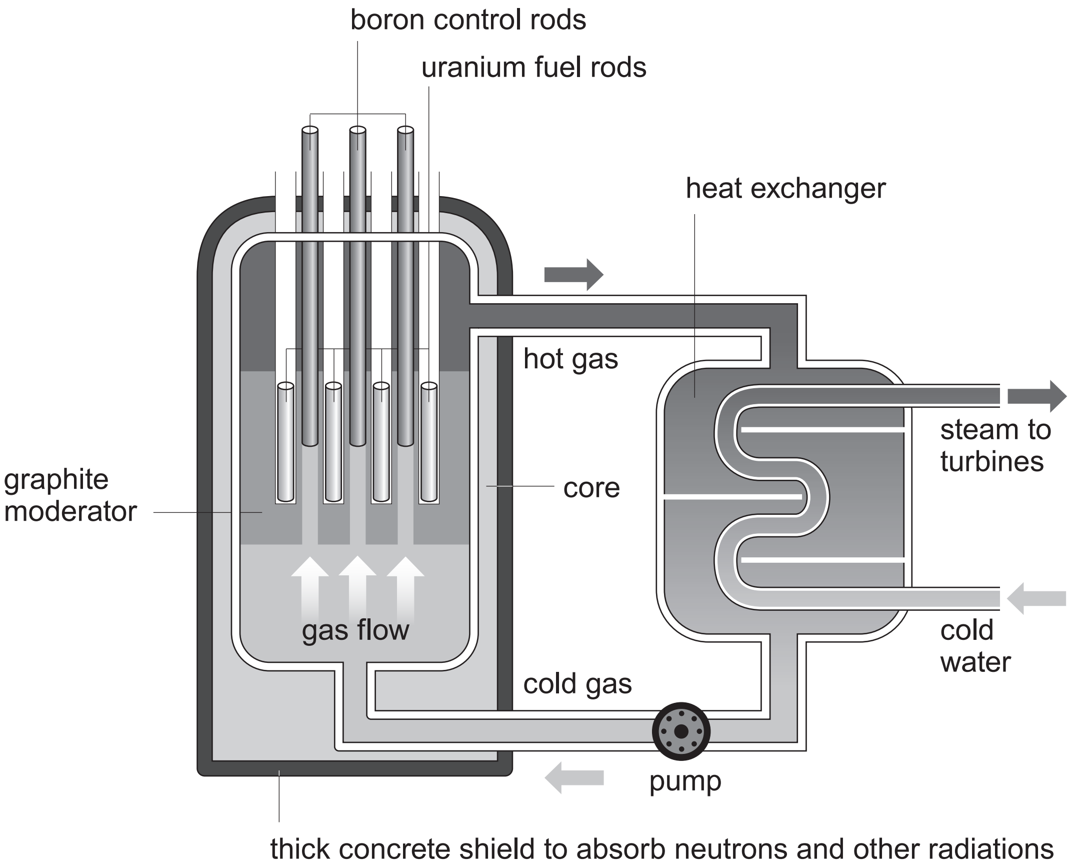
Examiner only

3. The diagram shows the structure of a nuclear reactor. The boron control rods and pump are powered by mains electricity.



Explain why the reactor cannot function safely during an interruption to the supply of electricity in a power cut. Explain how a nuclear accident is prevented and the consequences of a nuclear accident. [6 QER]

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Examiner only

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Examiner only

4. Colorimetry is a method used to determine the concentration of haemoglobin in blood.

The table below shows how the absorbance depends on the concentration of haemoglobin. One mean value is missing.

Concentration of haemoglobin ($mg/100\,cm^3$)	Absorbance (units)				
	Test 1	**Test 2**	**Test 3**	**Test 4**	**Mean**
50	0.20	0.21	0.20	0.19	0.20
100	0.37	0.41	0.63	0.39	
150	0.60	0.61	0.60	0.60	0.60
200	0.80	0.79	0.81	0.83	0.81
250	0.99	0.99	0.99	0.99	0.99

(a) Rhodri and Dafydd calculated the missing mean absorbance value. Rhodri calculated a mean value of 0.45 and Dafydd calculated a value of 0.39. Show how both values were obtained and explain which is the better method of calculating a mean. [3]

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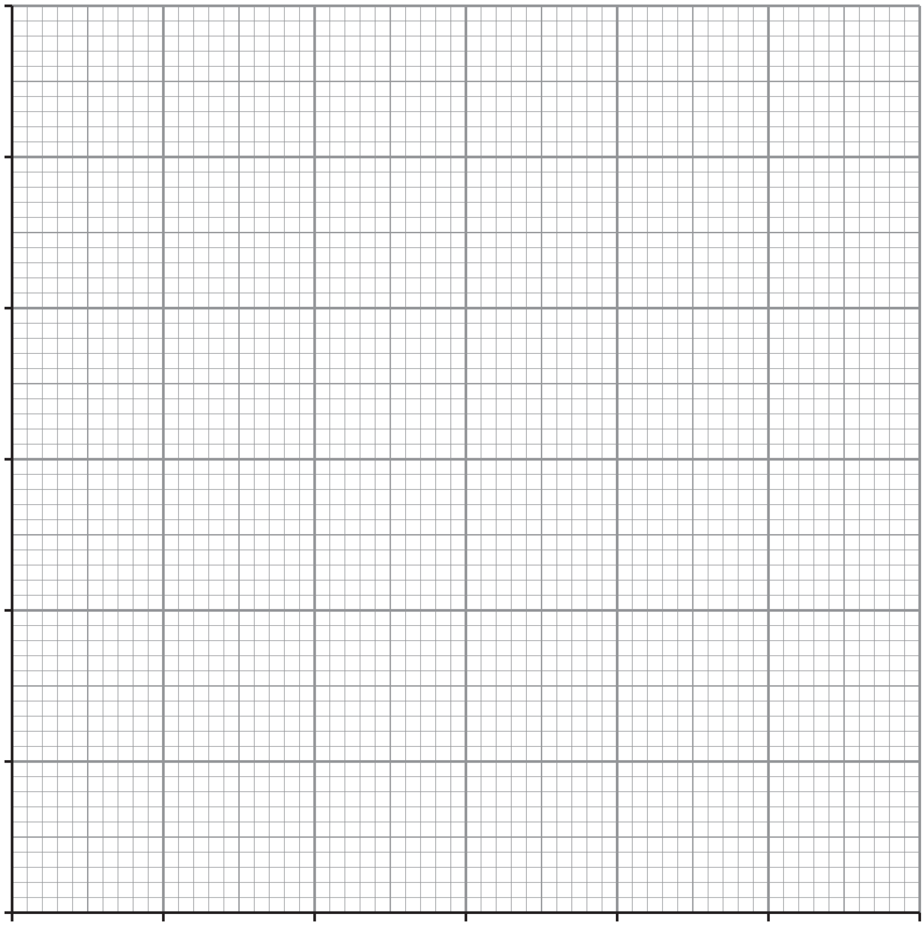
Examiner only

(b) (i) Use the data in the table to plot a graph of mean absorbance against concentration of haemoglobin on the grid below and draw a suitable line. [4]



(ii) Describe the relationship between absorbance and concentration of haemoglobin. [2]

....

....

(c) One part of the structure of haemoglobin has the chemical formula $FeN_4C_6H_{18}O_4$.

(i) Calculate the relative molecular mass (M_r) of this part of haemoglobin. [2]

M_r = ...

(ii) Calculate the total mass of this part of haemoglobin in 2.5 moles. [2]

Mass = ...

13

Examiner only

5. Hydrogen peroxide, H_2O_2, has anti-fungal properties and can be used to preserve milk. Before the milk is used to make cheese the H_2O_2 needs to be removed.

Hydrogen peroxide decomposes to form water and oxygen gas as shown below.

hydrogen peroxide $\longrightarrow$ water + oxygen

(a) Write a balanced symbol equation for the reaction. [2]

....

(b) The rate of this reaction is very slow at room temperature without a catalyst.

A student suggests that the rate of reaction can be increased by adding a catalyst or increasing the temperature.

State **two** variables that should be controlled in an experiment to determine the effect of temperature on the rate of decomposition of hydrogen peroxide. [2]

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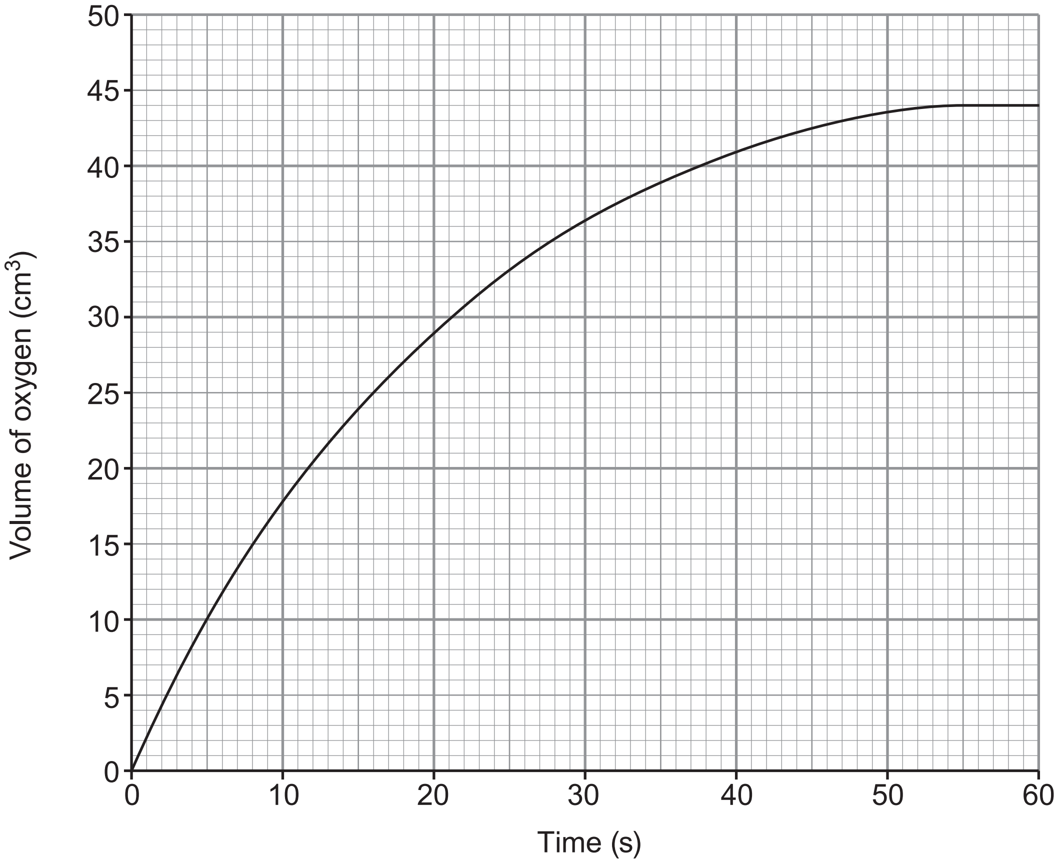
Examiner only

(c) In an investigation, the student measured the volume of oxygen collected in a measuring cylinder every 10 seconds for a total of 60 seconds.

The graph below shows the results at 22°C.



(i) Calculate the rate of reaction at 30 seconds by drawing a tangent. [3]

Rate = ... cm^3/s

(ii) The rate of reaction at 5 seconds was $2\,cm^3/s$. Explain, using particle theory, why the rate of decomposition was less at 30 seconds. [2]

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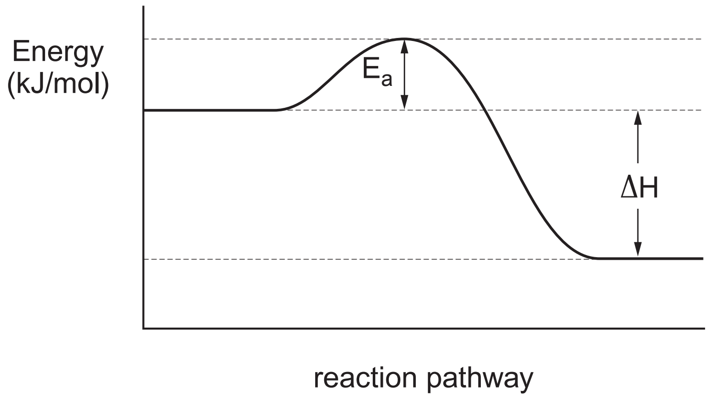
....

(iii) The experiment was repeated at 35°C. **Add** a line to the grid to show the results you would expect. [2]

Examiner only

(d) The energy changes during the decomposition of hydrogen peroxide without a catalyst are shown below. The energy E_a is 75 kJ/mol and ΔH is -196.1 kJ/mol.



(i) Explain what type of reaction is shown by this energy diagram. [2]

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(ii) A student states that the values of E_a and ΔH do not change when a catalyst is used. Explain whether you agree with the student. [3]

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(e) The rate of a chemical reaction doubles if the temperature is increased by 10°C.
At 6°C, milk undergoes a chemical reaction that makes it go sour in 6 days. Calculate how long it will take milk to go sour at 36°C. [2]

Time = ...

18

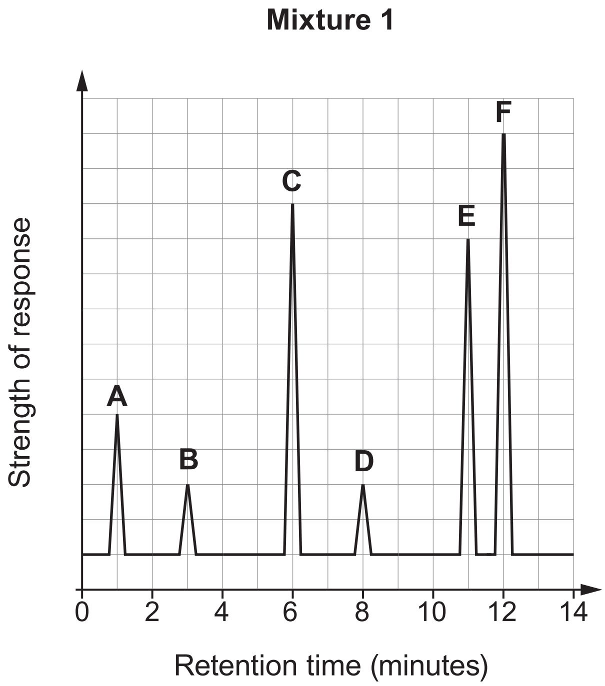
Examiner only

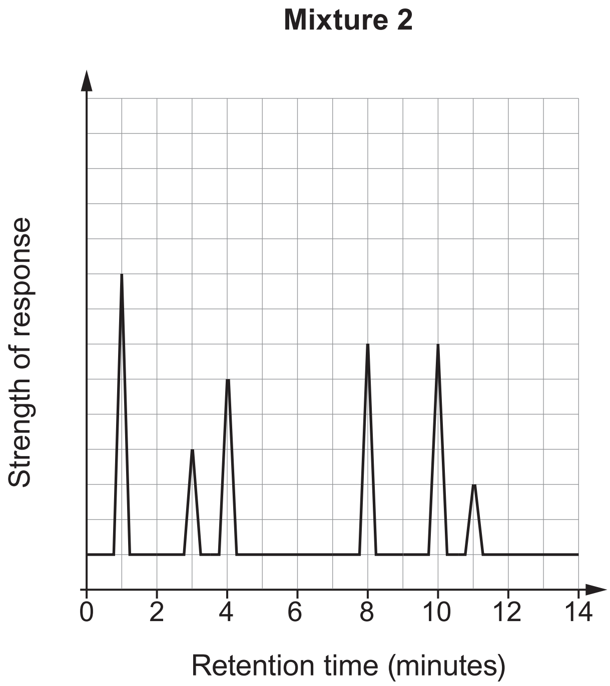
6. Chromatography is an important technique used by analytical chemists to separate substances found in mixtures.

Gas chromatography can be used to show:

- the number of compounds
- the identity of each compound
- how much of each compound is present in a mixture.

The diagrams below show the gas chromatograms from **mixtures 1** and **2**. Each mixture contains more than one compound. The compounds in **mixture 1** are labelled **A-F**.





(a) Use the chromatograms to compare the similarities and differences between **mixtures 1** and **2**. [4]

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(b) Explain in terms of molecules why **F** has the greatest retention time. [2]

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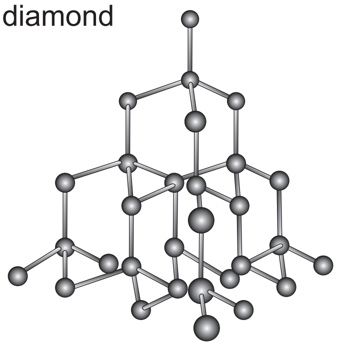
6

Examiner only

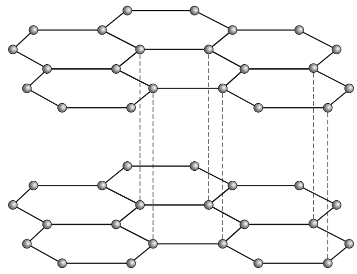
7. Allotropes of carbon include diamond, graphite, graphene, carbon fibres, fullerenes and carbon nanotubes.

(a) The structures of diamond and graphite are shown below.

diamond



graphite



Answer the following questions with reference to the bonding in diamond and graphite.

(i) Explain why diamond is an insulator of electricity but graphite is a conductor. [4]

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(ii) Explain why graphite is soft and slippery but diamond is hard. [3]

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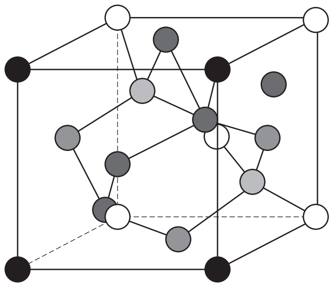
Examiner only

(b) A unit cubic cell of diamond is shown below.



Each side has dimensions of 0.36 nm (1 nm = 10^{-9} m).

The number of carbon atoms in a cubic cell is 8.

The mass of a carbon atom is 2×10^{-26} kg.

Use the information above and the equation:

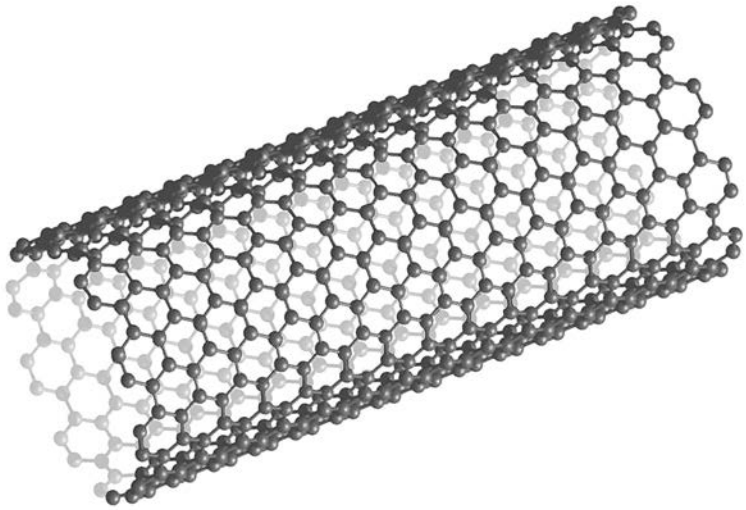
$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the density of this unit cubic cell of diamond to 2 significant figures. [4]

Density = ... kg/m^3

Examiner only

(c) A carbon nanotube is shown below.



Some properties of these nanotubes are compared with three other materials in the table below.

Material	Tensile strength (MPa)	Density (g/cm^3)
stainless steel	505	8.00
aluminium	572	2.81
Kevlar	3620	1.44
carbon nanotube	62000	1.34

Explain why carbon nanotubes are likely to replace materials such as steel and aluminium in the manufacture of power lines. [2]

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13

END OF PAPER

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3445UC0-1

Z22-3445UC0-1

TUESDAY, 17 MAY 2022 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes

HIGHER TIER

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**19**	
2.	**8**	
3.	**8**	
4.	**16**	
5.	**13**	
6.	**11**	
Total	**75**	

3445UC01
01

ADDITIONAL MATERIALS

A calculator.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **5**(c) is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of the examination paper.

Examiner only

Answer **all** questions in the spaces provided.

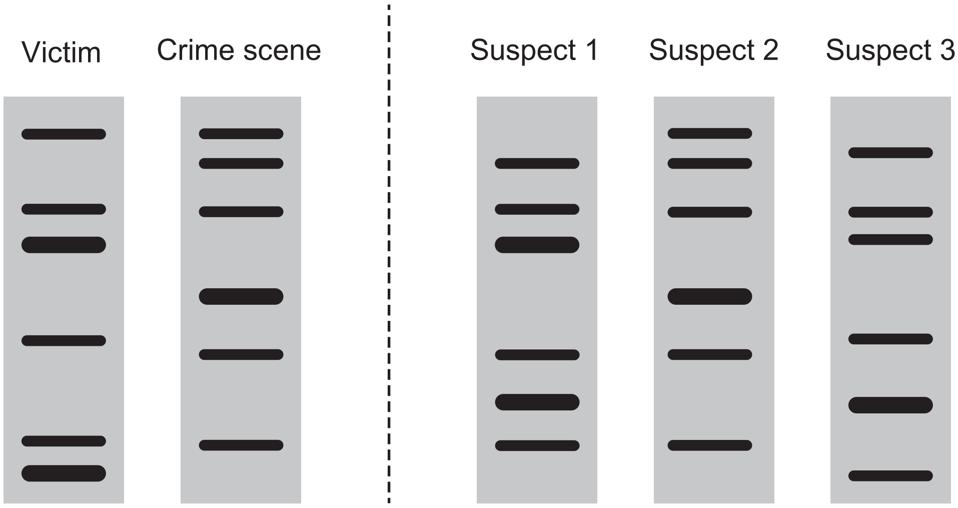
1. Criminals will try to hide their activities but forensic scientists, using biological and chemical testing, are able to find evidence that reveals what happened.

(a) DNA profiling is commonly used to identify criminals.

(i) State where DNA is found in the cell. [1]

....

(ii) There were three suspects in a crime. Suspect 1 did not have an alibi but suspects 2 and 3 did. Initially it was thought that suspect 1 was the criminal. DNA samples were collected from the crime scene.



Compare the DNA samples above to explain whether having an alibi or not is sufficient to decide on guilt. [2]

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(iii) State **one other** use of DNA profiling. [1]

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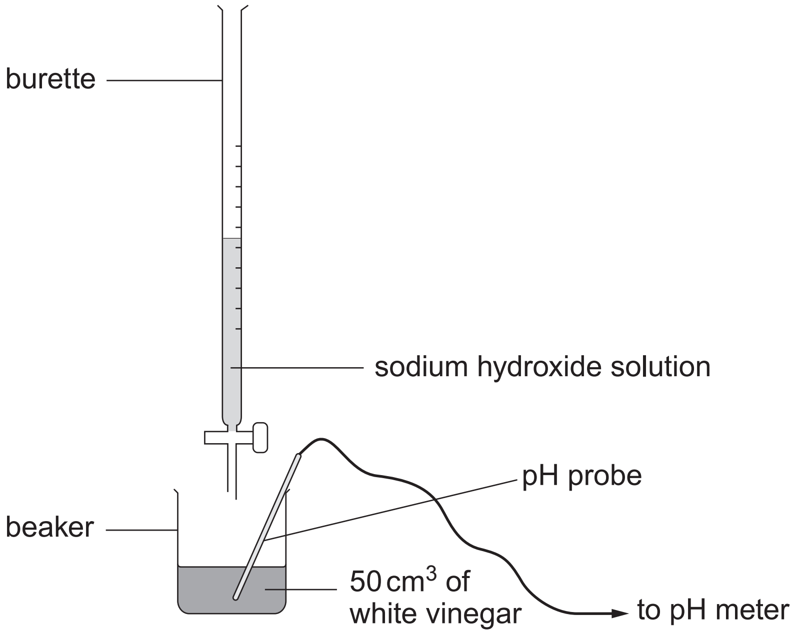
3445UC01
03

(b) It is suspected that supplies of vinegar have been tampered with by adding a strong acid. White vinegar usually contains ethanoic acid at a concentration of 0.01 mol/dm^3. Samples of the vinegar are taken and tested by titrating against 0.5 mol/dm^3 sodium hydroxide solution using the apparatus below.

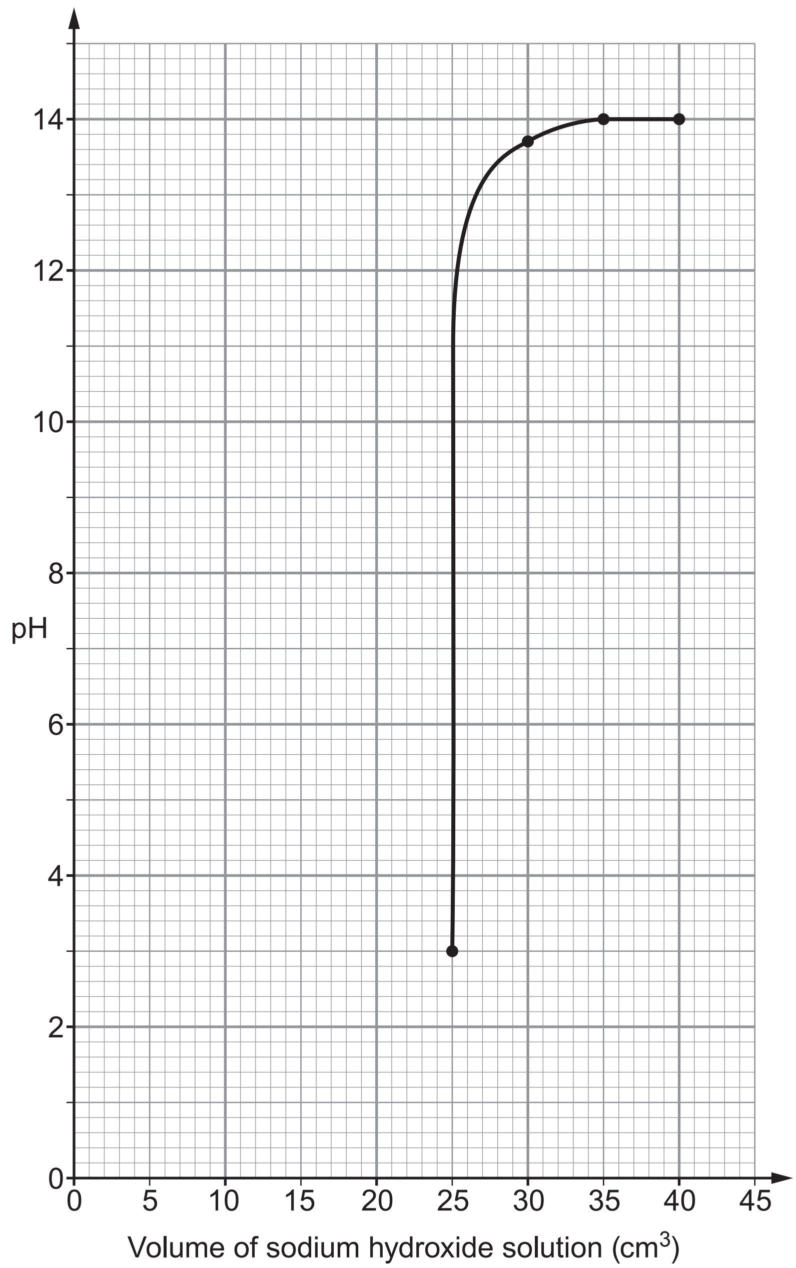


A pH meter is used to monitor the pH of the solution in the flask. The results are shown in the table below.

Volume of sodium hydroxide solution (cm^3)	pH reading
0	1.5
5	1.6
10	1.7
15	1.9
20	2.2
25	3.0
30	13.7
35	14.0
40	14.0

Examiner only

(i) Use the data in the table to complete the graph on the grid below. [3]



(ii) Use the graph to find the volume of sodium hydroxide solution required to neutralise the white vinegar. [1]

volume = ... cm^3

3445UC01 05

Examiner only

(iii) Use the information on pages 4 and 5 and the equation

$$\text{concentration of white vinegar} = \frac{\text{concentration of sodium hydroxide} \times \text{volume of sodium hydroxide}}{\text{volume of white vinegar}}$$

to determine whether or not the samples had been tampered with. [3]

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(iv) Complete the word equation for the reaction between an acid and a base. [2]

acid + base → ....

(v) Explain why taking multiple readings will increase the accuracy of the result. [2]

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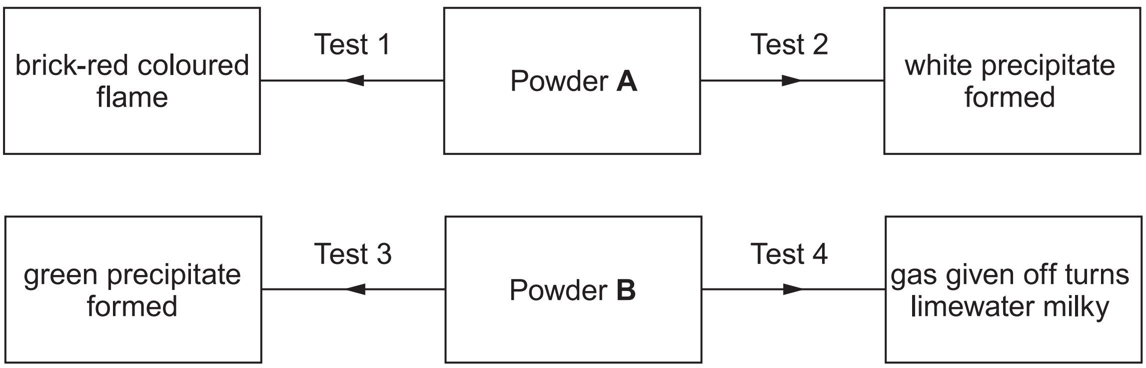
Examiner only

(c) Powders **A** and **B** are collected at a crime scene. The following tests are used to identify them.

Test	Action
1	Flame test
2	Add silver nitrate solution
3	Add sodium hydroxide solution
4	Add dilute hydrochloric acid and bubble gas through limewater

The results are given below.



Identify powders **A** and **B**. [4]

Powder **A** ....

Powder **B** ....

3445UC01 07

19

Examiner only

2. Microorganisms are regularly used in food production. However, unless precautions are taken, microorganisms can also cause food poisoning.

(a) (i) Describe the process of making cheese from milk. [3]

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(ii) State the optimum conditions required in this process. [3]

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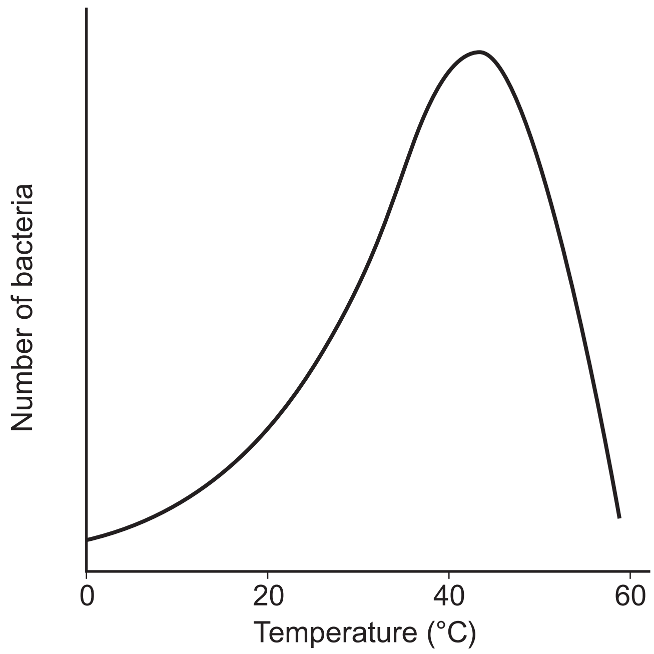
(b) Pasteurisation is a process that allows milk to be stored for longer.
The graph shows how bacterial growth varies with temperature.



Use the information in the graph to explain why milk is pasteurised at temperatures of about 70 °C. [2]

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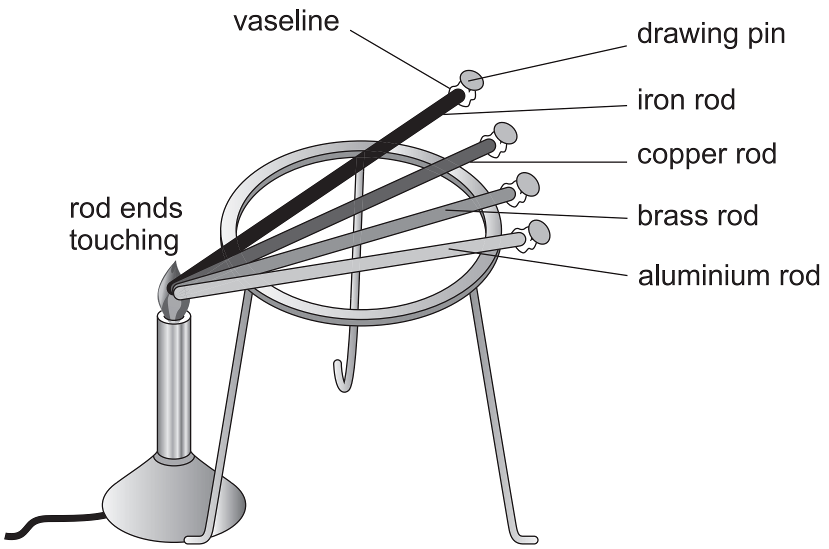
8

Examiner only

3. Metals are good conductors of heat and electricity.

(a) Three students investigated the thermal conductivity of four metal rods as shown below.



State **two** ways in which the students will make the experimental method valid. [2]

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(b) The table shows results from the students' investigation together with theoretical values of thermal conductivity.

Metal	Time taken for pin to drop (s)	Thermal conductivity (W/m K)
iron	17	80.4
copper	7	401
brass	12	109
aluminium	5	210

Use the data to explain whether the students' results are as expected. [3]

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Examiner only

(c) Explain, in terms of structure and bonding, why the electrical conductivity of sodium chloride is dependent upon its state. [3]

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8

4. Food self-sufficiency means that a country produces enough food to feed all of its people without having to import food from other countries.

(a) Discuss the impact of using fertilisers **and** pesticides in farming. [4]

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(b) Students investigated the effect of one type of NPK fertiliser.

Rice plants were treated with solutions of different concentrations of this NPK fertiliser. They were treated each day for five weeks.

The growth of each plant was recorded and the results are shown in the table below.

Concentrations of NPK fertiliser (units)	Plant growth after 5 weeks (mm)				
	plant 1	plant 2	plant 3	plant 4	mean
0.2	27	29	32	28	29
0.4	55	57	58	62	58
0.6	65	67	68	72	68
0.8	70	72	73	77	73

Examiner only

(i) Calculate the difference in mean plant growth **per week** for the plants treated with 0.2 and 0.6 units of NPK fertiliser. [3]

difference = ... mm/week

(ii) John suggests that mean growth **always** doubles if the concentration of fertiliser is doubled. Use data from the table to explain whether you agree with him. [3]

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(c) Describe how plant breeding programmes are used to produce crops with higher rice yields. [4]

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(d) Hydroponics is used to grow plants under controlled conditions.
Describe how hydroponics differs from traditional methods of growing plants. [2]

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16

Examiner only

5. The release of a cloud of highly toxic methyl isocyanate **(MIC)** gas from a pesticide plant in Bhopal led to the deaths of thousands of people.
On the night of the disaster **MIC** reacted with water that leaked into a storage tank, causing an explosion. As a result, **MIC** was released into the atmosphere.

The chemical formula of MIC is CH_3NCO and the equation for the reaction of MIC with water is shown below.

$$2CH_3NCO + H_2O \rightarrow C_3H_8N_2O + CO_2$$

(a) Calculate the relative molecular mass *(M_r)* of MIC using information from the Periodic Table on page 16. Show your workings, starting with the relative atomic mass (A_r) of each element found in **MIC**. [4]

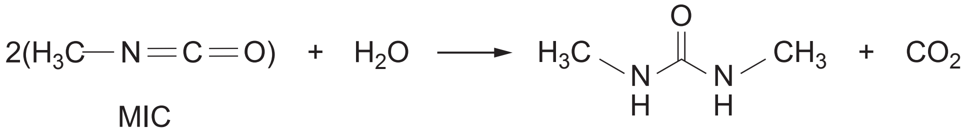
M_r = ...

(b) Approximately 30 000 kg of MIC gas was released. Calculate the number of moles released to three significant figures using your answer in (a). [3]

number of moles = ...

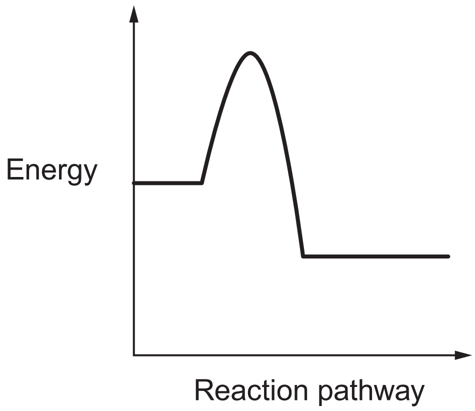
Examiner only

(c) The structural equation and reaction pathway for the reaction causing the explosion in the Bhopal plant are shown below.



MIC

1,3–dimethylurea



Use **all** the information above to explain why the reaction between MIC and water resulted in an explosion in terms of reaction pathway, bond energies and thermal runaway. [6 QER]

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13

Examiner only

6. Modern javelins are made of aluminium alloys or aluminium composite materials.

(a) Describe what is meant by the term 'composite material'. [2]

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(b) Pure aluminium is too flexible to make javelins. Copper can be mixed with aluminium to produce an alloy.

Pure aluminium   Copper and aluminium alloy

Use the diagrams and your own knowledge of alloys to explain how the presence of copper atoms in the aluminium results in the alloy being more rigid than pure aluminium. [3]

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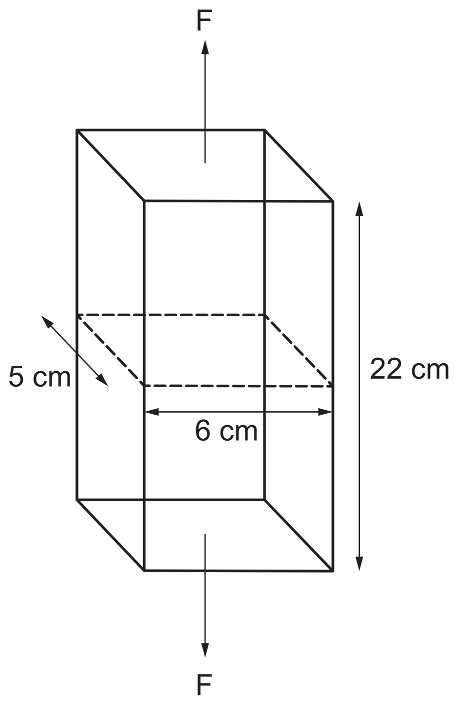
Examiner only

(c) An engineer was asked to determine the force required to break a pure aluminium rectangular bar.



The tensile strength of pure aluminium is 90 MN/m^2.

Use the information above and the equation

$$\text{tensile strength} = \frac{\text{force}}{\text{cross-sectional area}}$$

to calculate the force (F) required to break this bar. [6]

(1 MN = 10^6 N and 1 cm^2 = 10^{-4} m^2).

force (F) = ... N

11

END OF PAPER

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3445UC0-1

S23-3445UC0-1

TUESDAY, 16 MAY 2023 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 3: Food, Materials and Processes

HIGHER TIER

1 hour 30 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**14**	
3.	**10**	
4.	**13**	
5.	**13**	
6.	**12**	
7.	**8**	
Total	**75**	

3445UC01 01

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3**(b) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

The Periodic Table is printed on the back cover of the examination paper.

VP*(X21-3445UC0-1)

Examiner only

Answer **all** questions in the spaces provided.

1. (a) State, in terms of electrons, the difference between ionic and covalent bonding. [2]

....

....

....

(b) When magnesium reacts with oxygen, the ionic compound magnesium oxide is formed.

(i) The outer electrons of magnesium and oxygen are shown below. Draw arrows to show the transfer of electrons when magnesium oxide is formed. [1]

(ii) State the charges on the ions formed [2]

Mg .... O ....

3445UC01
03

5

Examiner only

2. Inos is a Welsh caravan manufacturer based in Denbighshire.
The materials used to build modern caravans are very different to those used in early caravans.

Table 1 shows how the materials used in manufacturing the body of a caravan have changed over time.

Year	Materials used
1920	wood
1950	steel and wood
1970	aluminium and wood
2015	glass reinforced plastic (GRP), polyester

Table 2 shows information about some of the materials used to make the body of caravans.

Material		Density (g/cm^3)	Stiffness (GPa)	Melting Point (°C)	Strength (MPa)	Is it brittle?	Does it corrode?
wood		0.6			1000	No	no but rots when wet
steel		7.8	210	1357	1200	No	yes producing rust
aluminium		2.7	69	660	90	No	yes producing a resistant coating
GRP	glass fibres	2.4	180	1400	3500	Yes	No
	epoxy resin	1.5			60	Yes	No
polyester		1.9	150	121	250	Yes	No

Use the information above to answer the following questions.

(a) Describe how the weight and strength of caravan bodies changed between 1920 and 1970. [4]

Weight: ....

....

....

Strength: ....

....

....

 (3445UC0-1)

Examiner only

(b) (i) State **two** advantages of the materials used in 2015 over those used previously. [2]

1. ....

2. ....

(ii) State **one** disadvantage of the materials used in 2015 over those used previously. [1]

....

(c) GRP is a common composite material, in which glass fibres are mixed with epoxy resin.

(i) A GRP panel contains a volume of $300\,\text{cm}^3$ of epoxy resin.

Use the equation: [3]

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the resin.

3445UC01 05

mass = ... g

(ii) I. State the strength of glass fibre in N/cm^2. [2]

$$(1\,\text{MPa} = 100\ \text{N/cm}^2)$$

strength = ... N/cm^2

II. The cross-sectional area (csa) of a glass fibre is $0.00015\ \text{cm}^2$.

Use your answer in part (c)(ii)I. and the equation:

$$\text{force} = \text{strength} \times \text{csa}$$

to calculate the force required to break it. [2]

force = ... N

14

Examiner only

3. Ash dieback disease is a fungal infection caused by a fungus called *Hymenoscyphus fraxineus* that is common in ash trees in Wales.

The disease causes the leaves to blacken and wilt, so eventually the ash tree has no leaves.

The photographs show healthy ash leaves and those affected by ash dieback.

(a) Describe the process of photosynthesis **and** explain why dieback disease affects the growth of the ash tree. [4]

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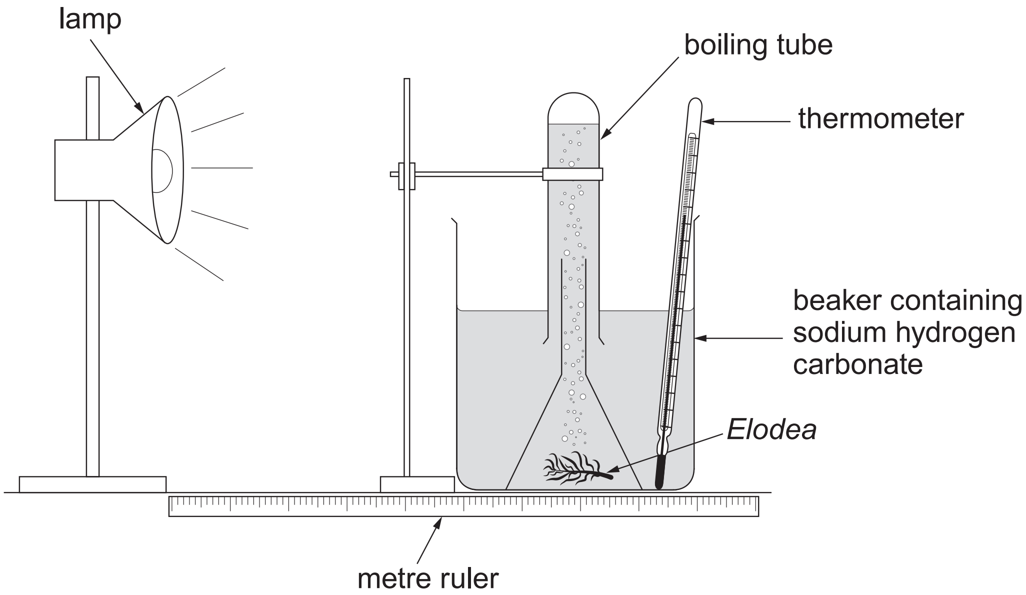
Examiner only

(b) Describe an experiment to investigate how limiting factors affect the rate of photosynthesis in the *Elodea* plant. [6 QER]



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3445UC01 07

10

Examiner only

4. Nuclear reactor 3 at the Hunterston B power station was shut down after cracks in graphite moderator bricks were found to be growing faster than expected. The Office for Nuclear Regulation (ONR) raised concerns over the number of cracks in the joints between the graphite moderator bricks in the core. An unusual event, such as an earthquake, might move the graphite bricks so the control rod channels could become blocked.

(a) (i) Explain the purpose of the graphite moderator in a nuclear reactor. [2]

....

....

....

(ii) Explain why blocked control rod channels are very dangerous. [3]

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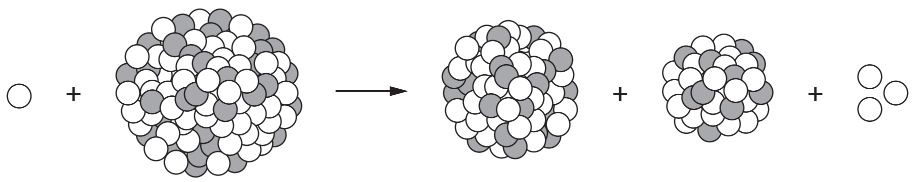
(b) The diagram shows a reaction in a nuclear reactor and part of the reaction equation.

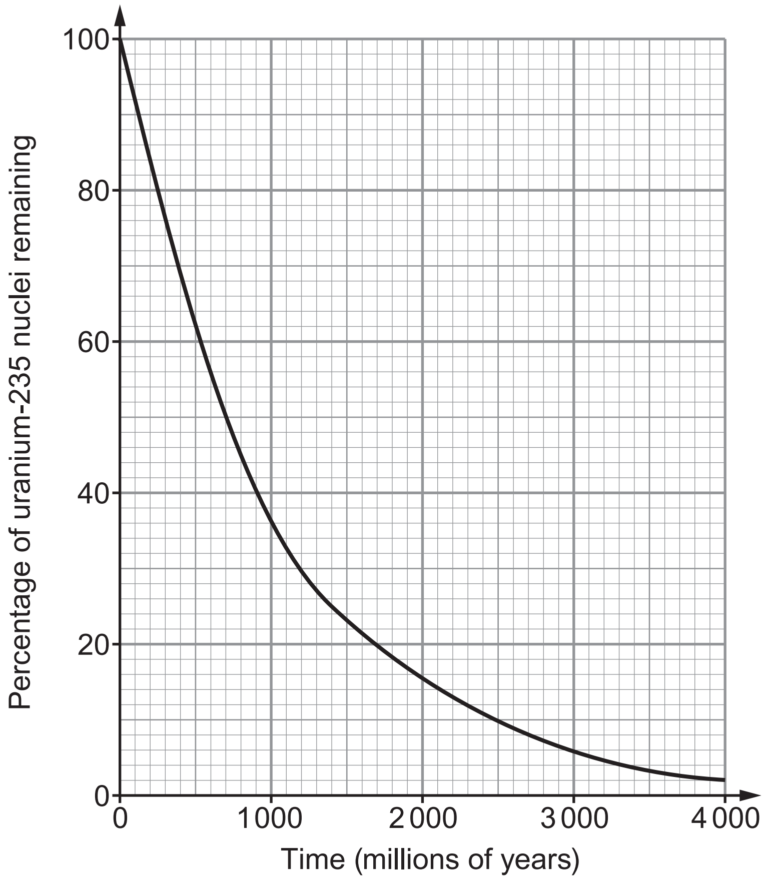


$$ {}^{1}_{0}\text{n} + {}^{235}_{92}\text{U} \longrightarrow {}^{141}_{56}\text{Ba} + {}^{\dots\dots}_{\dots\dots}\text{Kr} + \dots\dots\ {}^{1}_{0}\text{n} $$

Complete the equation. [3]

Examiner only

(c) It is likely that Hunterston B power station will be decommissioned. When it is, the uranium-235 in the fuel rods will continue to decay at the rate shown in the graph below.



(i) Add lines to the graph to calculate the half-life of uranium-235. [2]

half-life = ... million years

(ii) Uranium-235 becomes safe once its activity drops to 1/32 of its original value. Calculate the time it takes to drop to this value. [3]

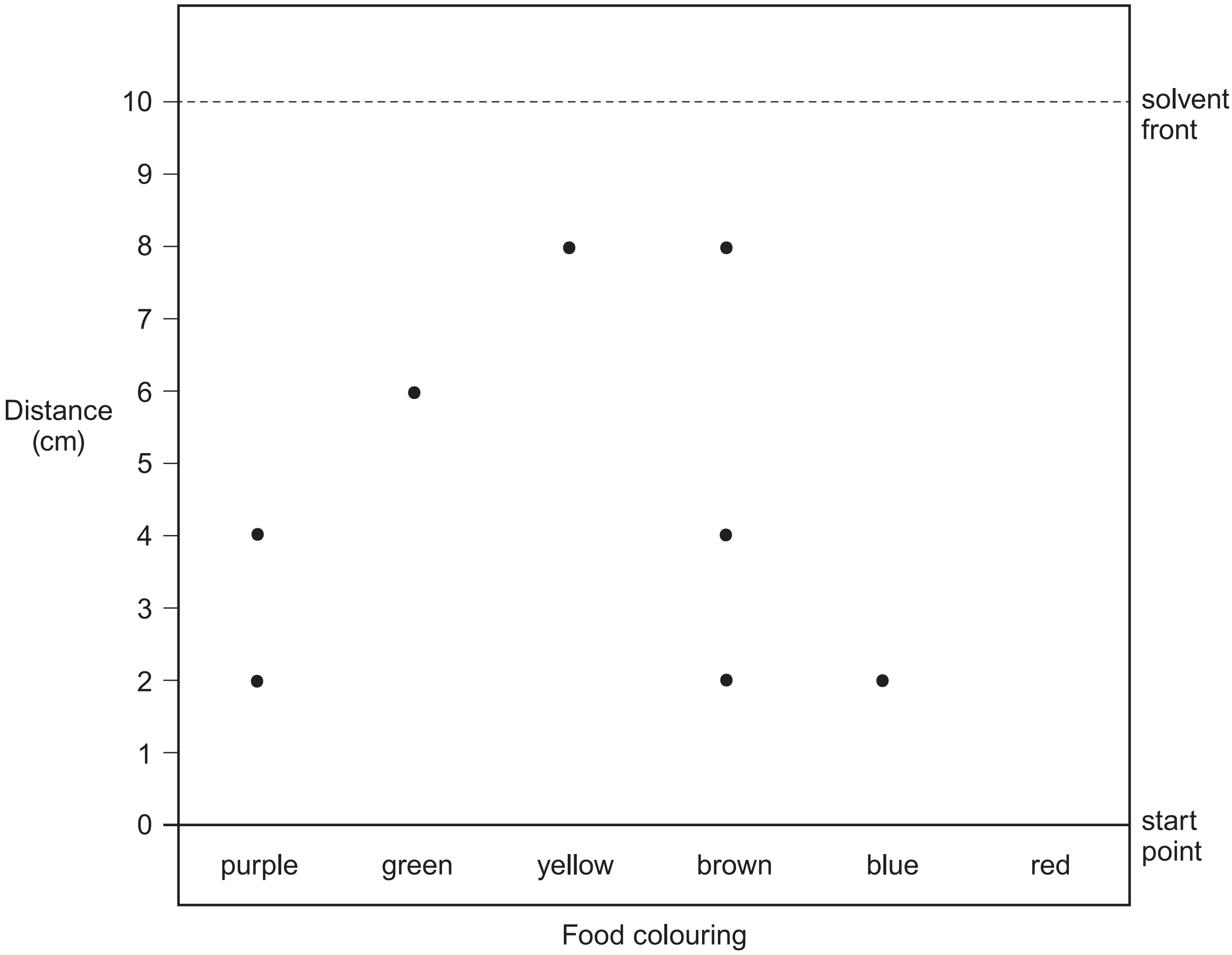
time = ... years

13

5. Analytical scientists work in a wide range of different industries and agencies.

(a) The Food Standards Agency (FSA) commissioned a project to detect the coloured dyes used in food colourings.

The diagram below shows a chromatogram of different food colourings.



Examiner only

(i) The result for the dye in the red food colouring is not included in the diagram. The R_f value for red colouring is 0.4.

Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

to calculate the distance moved by the dye in the red colouring on this chromatogram **and** add it to the diagram. [3]

distance travelled = ... cm

(ii) It was claimed that brown colouring was not a combination of other coloured dyes. Explain whether you agree with this claim. [2]

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(iii) Explain, in terms of the mobile and stationary phases, why the molecules in purple colouring move different distances along the chromatogram. [3]

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Examiner only

(b) Qualitative chemical testing is used to identify the ions in compounds.

(i) Six unlabelled bottles each contained a solution of one of the compounds in the box below.

copper(II) sulfate	sodium carbonate	lithium carbonate
sodium chloride	potassium sulfate	calcium iodide

I. State which compounds would produce a yellow flame in a flame test. [1]

....

II. State which compounds would produce bubbles of gas when added to hydrochloric acid. [1]

....

(ii) A technician has a colourless solution in an unlabelled bottle. He suspects it is metal chloride. Describe a test that could be carried out to confirm that the solution is a chloride, and give the expected result. [3]

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13

Examiner only

6. Levels of bacteria in milk are routinely monitored.

(a) Most milk sold to consumers is pasteurised and homogenised.

(i) Explain why milk is pasteurised. [2]

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(ii) Describe the process of homogenisation and explain why it is carried out. [4]

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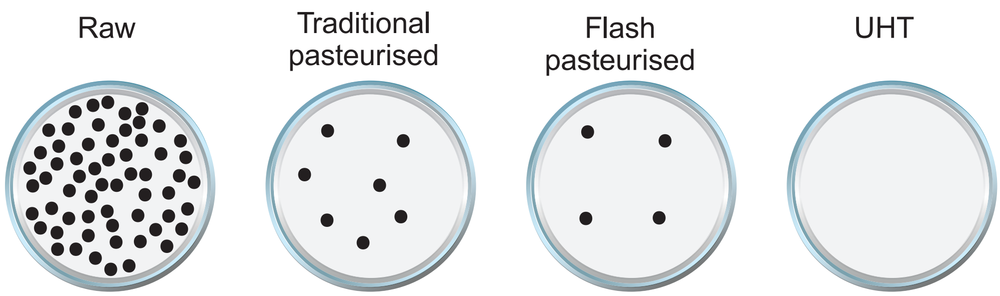
(b) Scientists investigated four types of milk. They monitored the growth of bacterial colonies on agar plates **using 0.1 cm³** from each type of milk over two days. Their results are shown below.



Type of milk	Heat treatment	Number of bacterial colonies in a 40 cm³ serving
raw	none	24 000
traditional pasteurised	63°C for 30 minutes	....
flash pasteurised	78°C for 35 seconds	1600
ultra-high temperature pasteurised (UHT)	135°C for 2 seconds	none

Examiner only

(i) Complete the table. [2]

Space for working

(ii) Unopened UHT milk can be stored on supermarket shelves at room temperature for six months but unopened pasteurised milk must be stored in a refrigerator at 4°C or below for no longer than seven days. Explain these differences. [4]

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Examiner only

7. A student investigated the reaction between hydrochloric acid (HCl) and calcium carbonate ($CaCO_3$). She measured the volume of gas produced for different concentrations of acid after 60 seconds.

She started each reaction at the same temperature, with the same mass of calcium carbonate and the same volume of acid. The acid was in excess each time.

The table below shows the time taken to produce 60 cm^3 of CO_2 for each concentration of HCl.

Concentration of HCl (mol/dm^3)	Time to produce 60 cm^3 of CO_2 (s)	Mean rate of reaction (cm^3/s)
0.5	234	0.26
1.0	118	0.51
2.0	58	....

(a) (i) **Complete** the table to two significant figures. [2]

(ii) Estimate the time taken to produce 60 cm^3 of carbon dioxide if a concentration of 4.0 mol/dm^3 HCl is used. [1]

time = ... s

Examiner only

(b) (i) State and explain, in terms of particles, the effect of increasing acid concentration on the rate of reaction. [3]

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(ii) The student noticed that when the experiments ran for 10 minutes, she obtained the same volume of carbon dioxide in each case.

Explain her observation. [2]

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8

END OF PAPER

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number